

Project 3

Computing in Context (SIPA)

Q Search



Syllabus

Course
info



Lectures



Labs



Reading



Projects



Homeworks
1-3



Project 1

Project 2

Project 3

Contents

- Objective
- Intro
- Process
- Prompt
- Analysis requirements

[General Project information](#)

Objective

Prove or disprove a hypothesis using skills learned in this class.

Intro

You can think of this as similar to the [Project 2](#) requirements, but expanded. [Examples of Final Projects for Python for Public Policy](#) - the result of

this Project will be similar.

Process

1. [Find a dataset](#) that seems interesting.
 - To meet the [requirement](#) that your project “not be trivial,” you probably want a dataset that’s large enough that you can’t understand it at a glance.
 - If you’re only using one dataset, you probably want it to have 500+ rows.
2. Load the data into a DataFrame.
3. Inspect the data a bit.
4. Fill out the prompt (below).
 - Work backwards:
On a piece of paper / whiteboard, draw the visualization you imagine

producing.

5. Use the data to answer the question.
6. If you end up answering your initial research question easily (haven't met [the requirements](#)), that's fine. Ask and answer follow-up question(s).
 - Go deep, not broad.

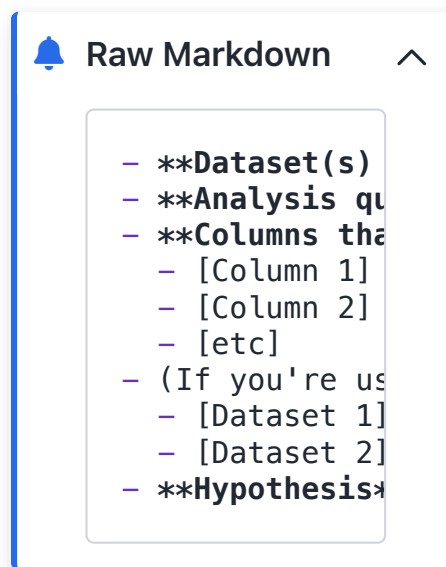
Prompt

Put the following in a Markdown cell in your notebook and fill it out:

- **Dataset(s) to be used:**
[link]
- **Analysis question:**
[question]
- **Columns that will (likely) be used:**
 - [Column 1]
 - [Column 2]
 - [etc]
- (If you're using multiple datasets) **Columns to**

be used to merge/join them:

- [Dataset 1]
[column]
- [Dataset 2]
[column]
- **Hypothesis:**
[hypothesis]



The question should be:

- Specific
- Objectively answerable through the data available
- *Just the right amount of ambitious* ([non-trivial](#))

If you want help/feedback, don't hesitate to ask on [Ed](#) or come to [office hours](#).

Tips

- Don't overthink it; getting up through filling out the prompt shouldn't take more than a few hours.
- Your question/hypothesis doesn't need to be something novel; confirming something you read / heard about is fine.
- The point of the prompt is to ensure you've dug into the data and that your project scope is reasonable. Think of it as a guide rather than something you're locked into.
- Even the question can bake in assumptions.
 - Example: "What ZIP code has the highest number of food poisoning cases?" assumes a relationship between food-borne illness and

geography.

- What assumptions does your question make?

Analysis requirements

Your submission must:

- **Meet the [general Project information](#)**
- **Not be trivial** (or -30 points) - requiring:
 - At least 40 lines of code to come to a conclusion
 - That code should be relevant to answering your question. In other words, having 40 lines of

```
print("hello world")
```

 wouldn't count.
 - If you meet all the other

requirements,
you will likely
be well over
this number.

- You can count them automatically using a tool like [tokei](#).
- Operations that aren't easily done in a spreadsheet.
- **Transform data** through [grouping](#), [merging](#), and/or [reshaping](#) of DataFrames (or -15 points)
- **Have a visualization** (chart or map) of some kind (or -5 points)
 - [General requirements](#)
- **Have the [prompt](#) filled out**

[Submit.](#)